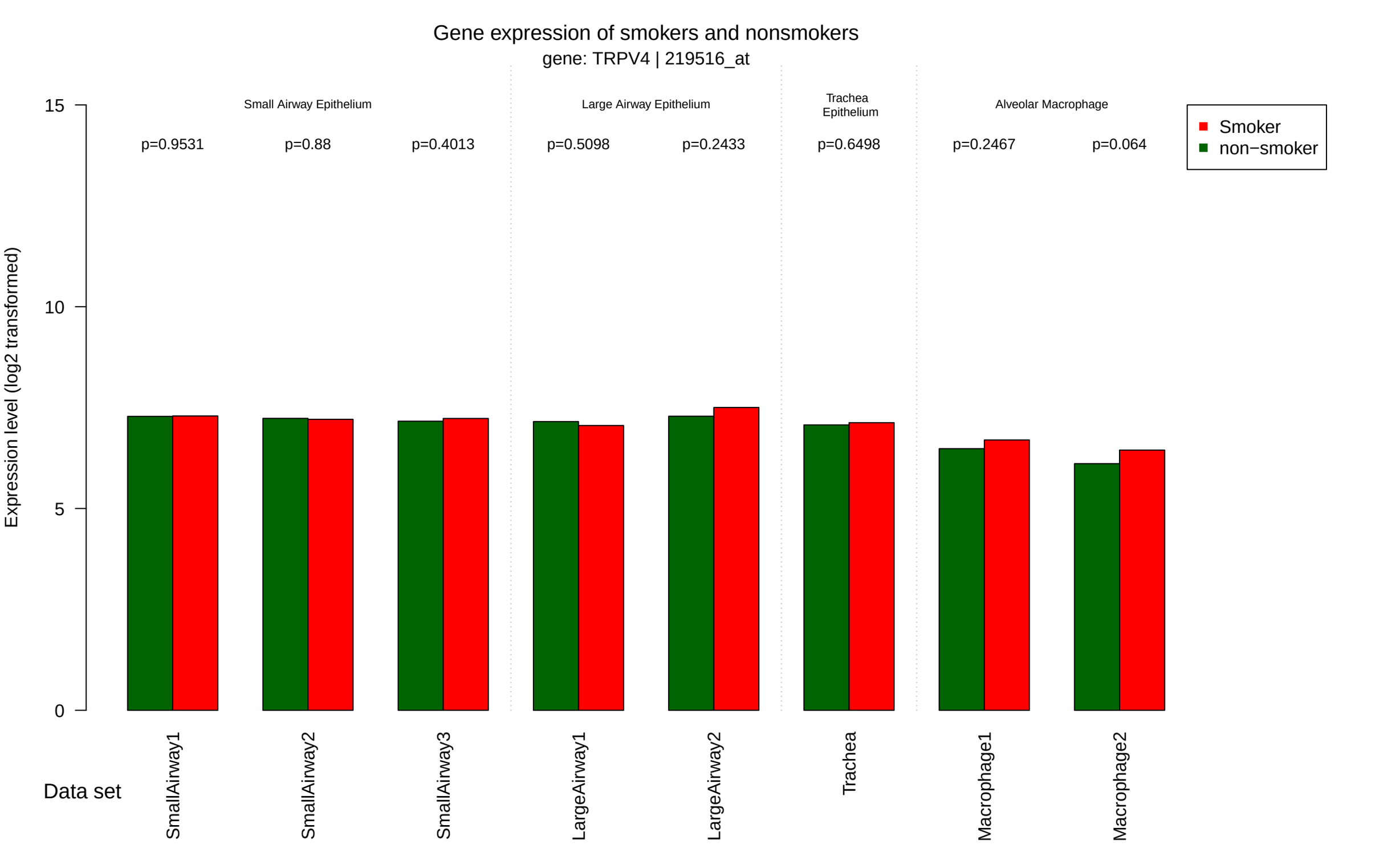
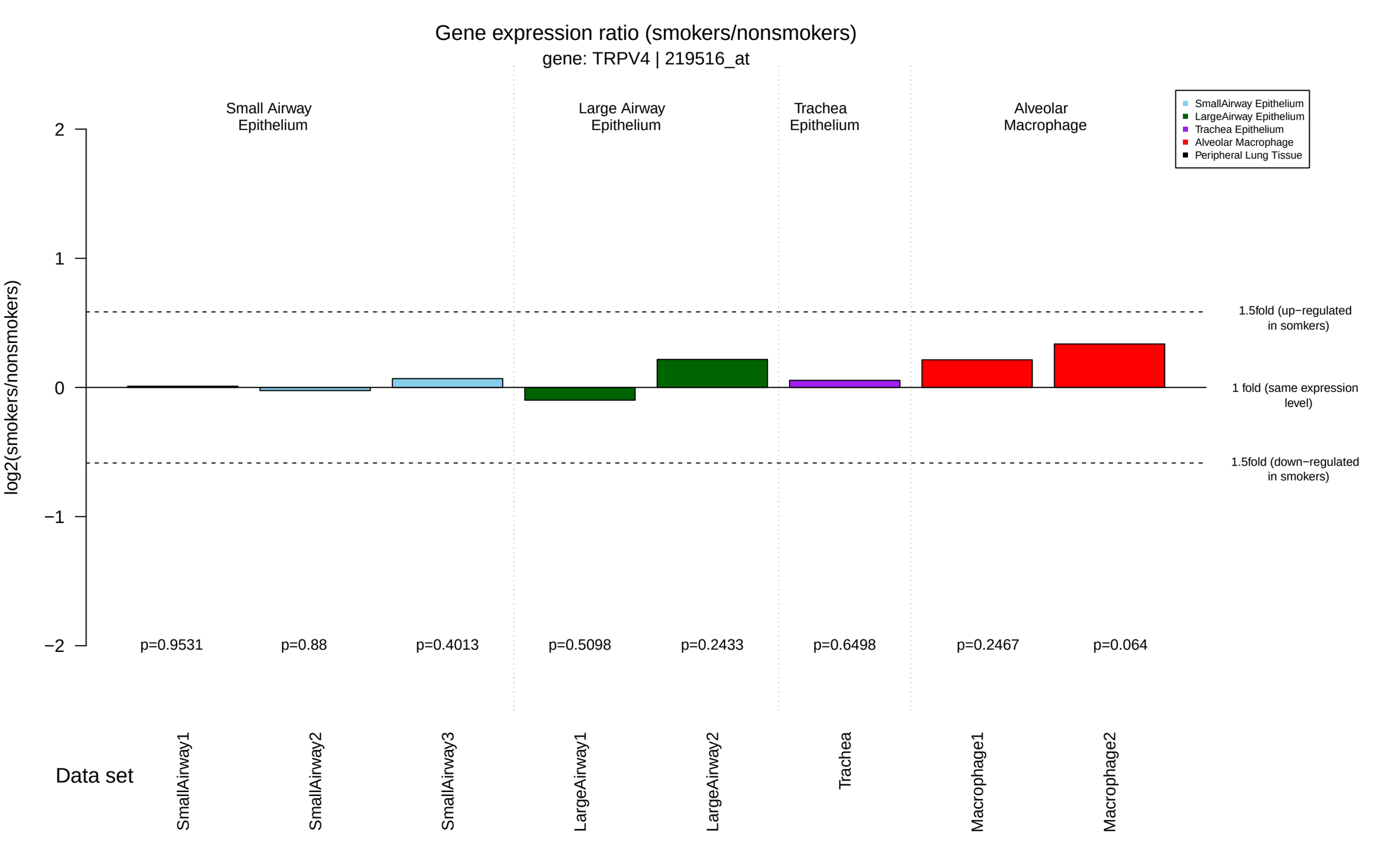
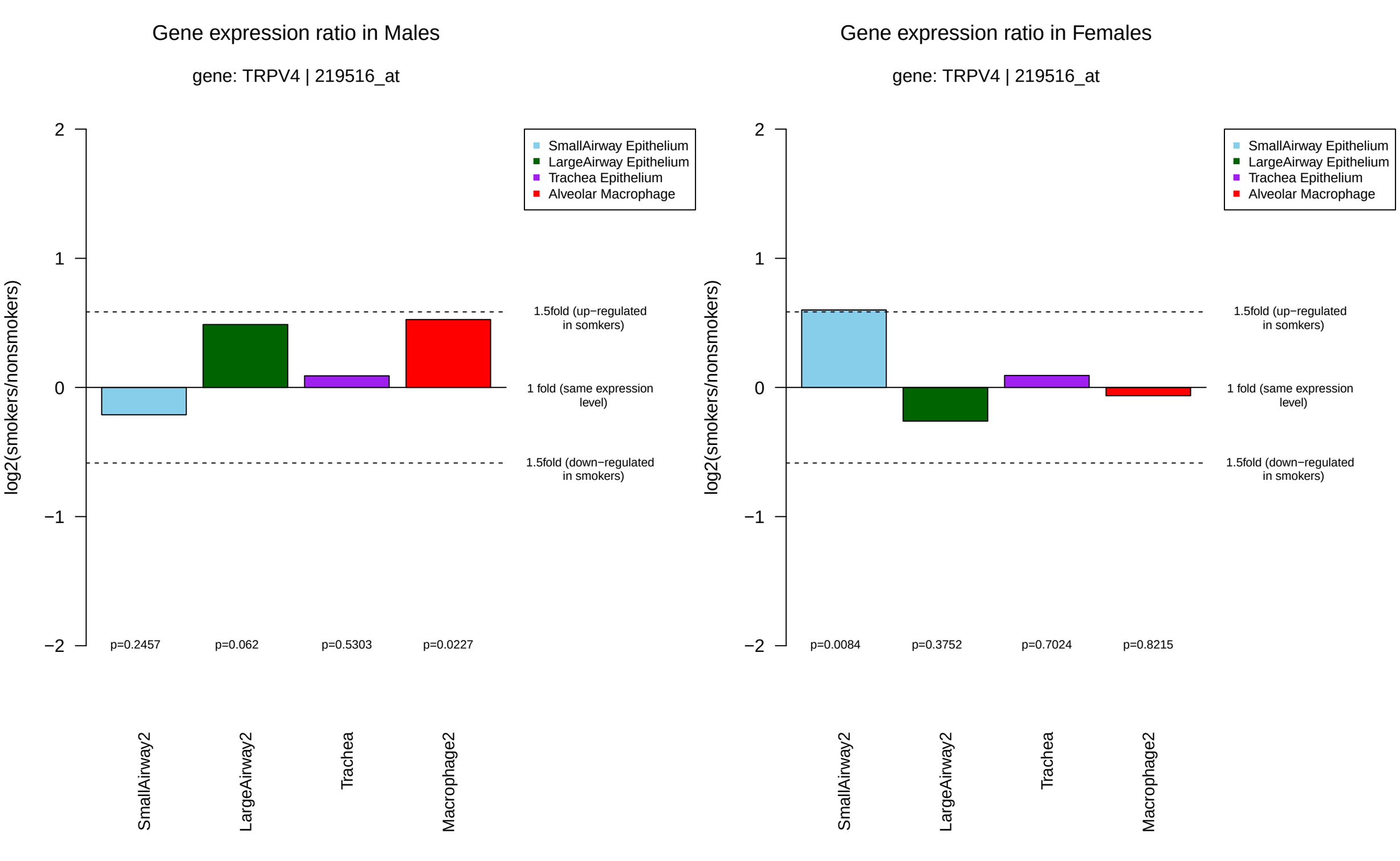
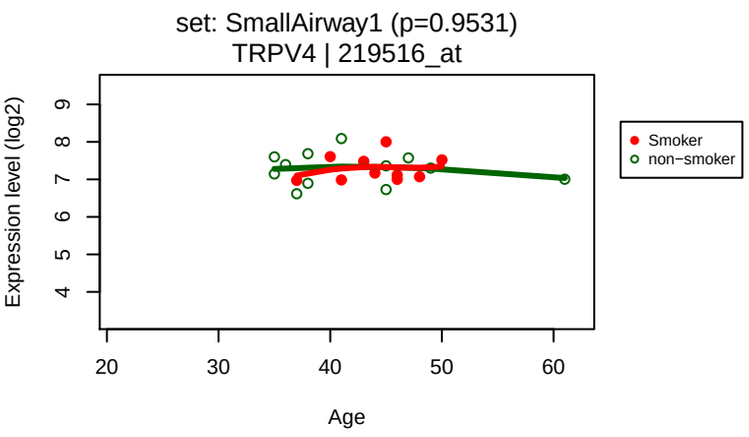
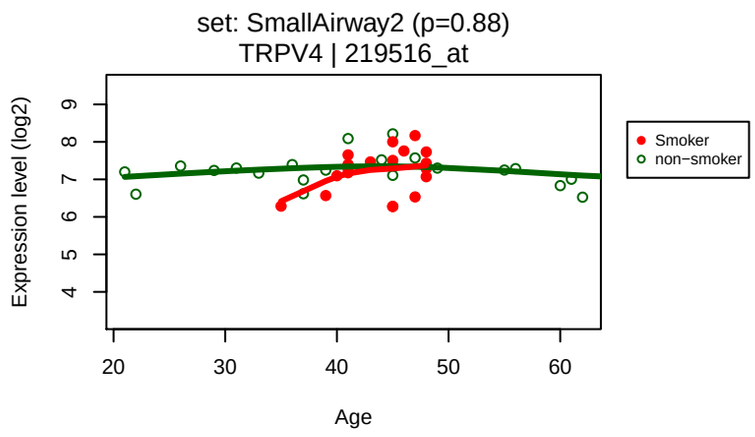
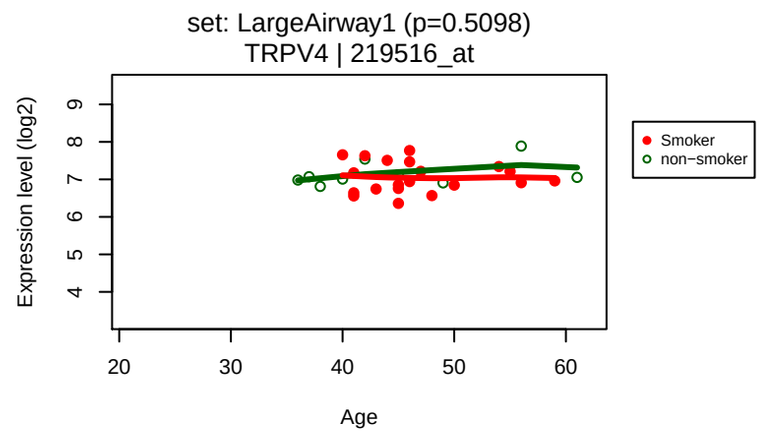

Gene expression of smokers and nonsmokers
gene: TRPV4 | 219516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9531
p=0.88
p=0.4013
p=0.5098
p=0.2433
p=0.6498
p=0.2467
p=0.064
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

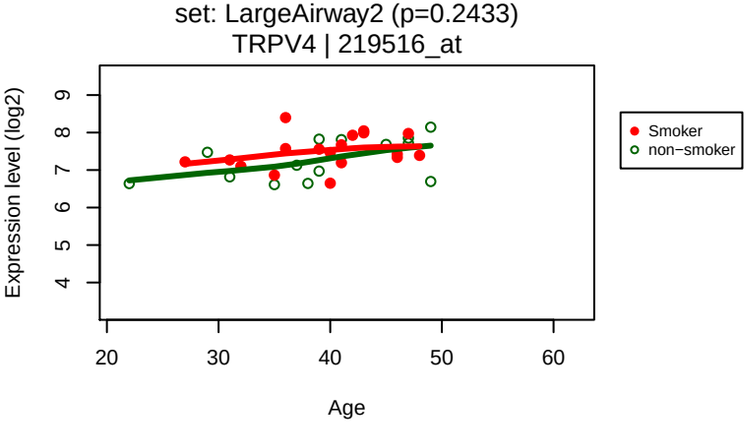

Gene expression ratio (smokers/nonsmokers)
gene: TRPV4 | 219516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9531
p=0.88
p=0.4013
p=0.5098
p=0.2433
p=0.6498
p=0.2467
p=0.064
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

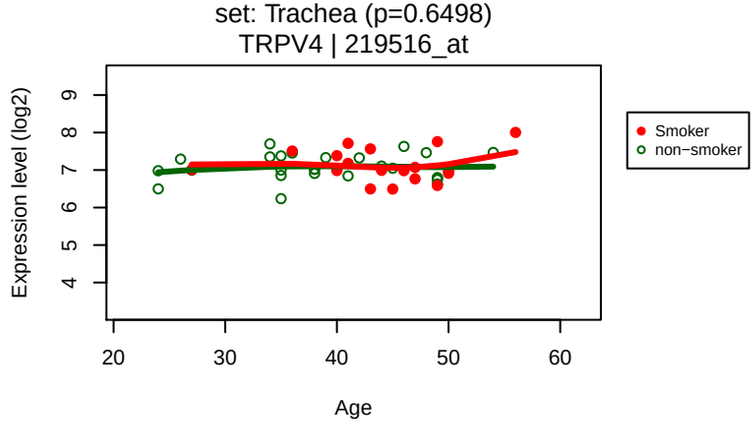

Gene expression ratio in Males
gene: TRPV4 | 219516_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2457
p=0.062
p=0.5303
p=0.0227
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRPV4 | 219516_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0084
p=0.3752
p=0.7024
p=0.8215
SmallAirway2
LargeAirway2
Trachea
Macrophage2

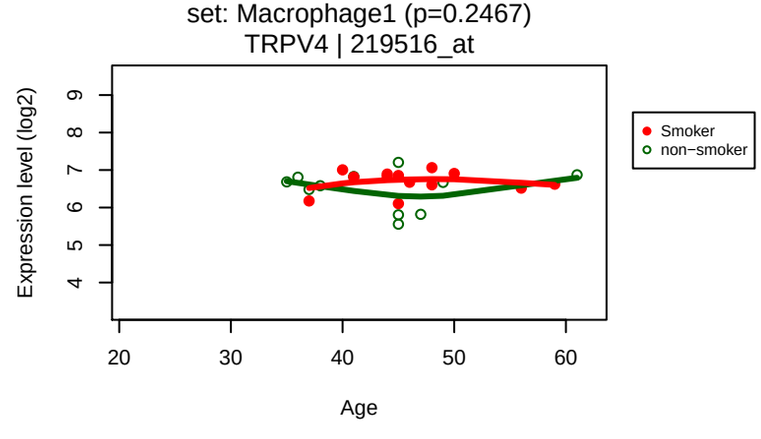

set: SmallAirway1 (p=0.9531)
TRPV4 | 219516_at
Expression level (log2)
Age
Smoker
non−smoker

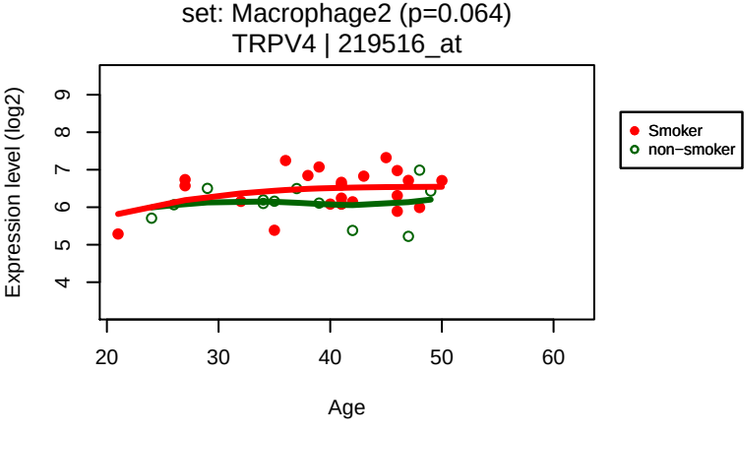

set: SmallAirway2 (p=0.88)
TRPV4 | 219516_at
Expression level (log2)
Age
Smoker
non−smoker

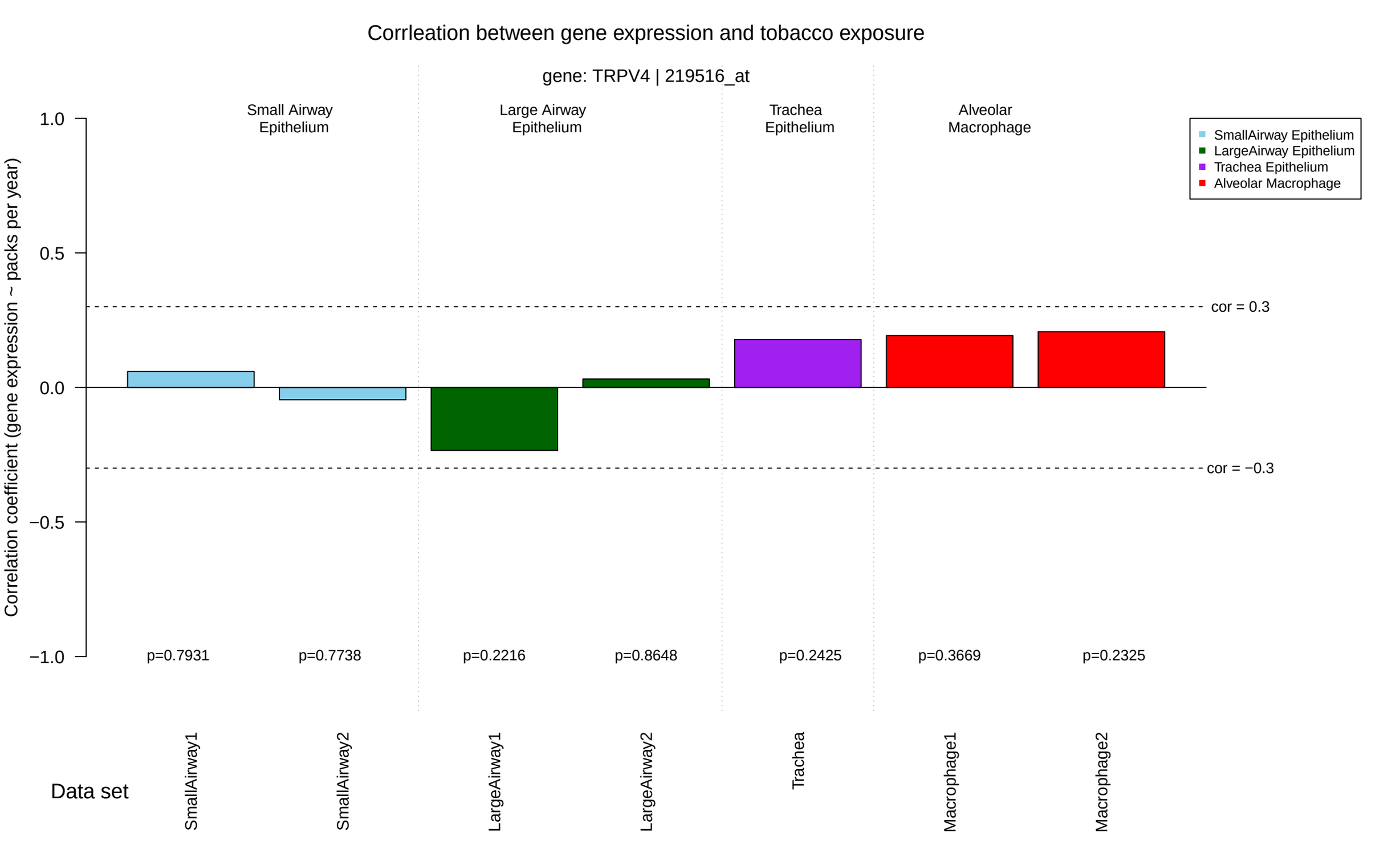

set: LargeAirway1 (p=0.5098)
TRPV4 | 219516_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2433)
TRPV4 | 219516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6498)
TRPV4 | 219516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2467)
TRPV4 | 219516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.064)
TRPV4 | 219516_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRPV4 | 219516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7931
p=0.7738
p=0.2216
p=0.8648
p=0.2425
p=0.3669
p=0.2325
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2